

NetMessenger Design Diary

Name: W.W.D.M. Fernando

Registration Number: IT23629844

Module: IE3010 - Network Programming

Project: NetMessenger: A Multi-Client Chat and File-Sharing Platform over TCP/IP

3 October 2026 – Initial Setup

- I reviewed the IE3010 Network Programming assignment requirements.
- I identified the required NetMessenger features.
- I calculated the personalized values using registration number IT23629844.
- The last four digits are 9844.
- Therefore:

Server Port = 6000 + 9844
= 15844

- Digits 3–6 of the numeric part 23629844 are 6298.
- Therefore:

Node ID = NID:6298

- I created the personalized source files:

server_9844.c

client_9844.c

Makefile_9844

- I created the README file and storage directory.
- I configured the server to use:

netmsg_IT23629844.log

- I initialized Git for the project.
- I later connected the local repository to the GitHub repository:

DMFernando0315/IT23629844-NetMessenger

4 October 2026 – TCP Client and Server Development

- I worked on the basic TCP client-server communication.
- I used C socket programming for communication between the client and server.
- I worked with functions such as socket(), bind(), listen(), accept(), connect(), send() and recv().
- I compiled the client and server using the personalized Makefile.
- I tested the connection between the client and server.
- I reviewed command processing and client connection handling.

5 October 2026 – User Registration and Messaging

- I implemented and tested user registration.
- I tested multiple usernames.
- I implemented the LIST command.
- I tested multiple clients connected to the server.
- I implemented broadcast messaging using BCAST.
- I implemented private messaging using PMSG.
- I verified that the server returned NID-tagged responses.
- I tested the system using several simultaneous clients.

5 October 2026 – Chat Rooms

- I implemented the chat-room functionality.
- I tested:

JOIN
ROOMS
RMSG
LEAVE

- I tested joining a room and sending messages to room members.
- I tested leaving a room.
- I checked the room list using ROOMS.

6 October 2026 – File Sharing

- I implemented the SENDFILE functionality.
- I created test.txt for testing.
- I sent the file from one client to another.
- I confirmed that the recipient received the file successfully.
- I checked the server storage directory.
- The required storage structure was:

./storage/IT23629844/<sender_username>/<filename>

6 October 2026 – Logging and Error Testing

- I checked the server log file:

netmsg_IT23629844.log

- I tested invalid operations and error responses.
- I tested a private message to an unavailable user.
- I tested an invalid command.
- I checked that the server continued running after invalid requests.

7 October 2026 – Disconnect and GitHub

- I tested QUIT.
- I disconnected and reconnected clients.
- I checked that the server continued running.
- I initialized the Git repository.
- I created the initial Git commit.
- I connected the repository to GitHub.
- I pushed the main branch successfully

Final Development Status

The main NetMessenger functionality was implemented and tested, including:

REGISTER
LIST
BCAST
PMSG
JOIN
LEAVE
ROOMS
RMSG
SENDFILE
QUIT